



Displaying Data from Multiple Tables using Joins

Objectives

After completing this lecture, you should be able to do the following:

Write `SELECT` statements to access data from more than one table using equality and nonequality joins by using Oracle Syntax.

View data that generally does not meet a join condition by using outer joins Oracle Syntax

Join a table to itself by using a self join Oracle Syntax

What is Join?

Joins are used to get data from more than one table.

To join more than one table we need at least one column common in both tables.

Tables get joined based on the condition specified. “ON” keyword is used.

There are different types of joins in SQL as below.

INNER Join

LEFT Join

RIGHT Join

FULL Join-*

OBTAINING DATA FROM MULTIPLE TABLES

EMPLOYEE_ID	FIRST_NAME	LAST_NAME	DEPARTMENT_ID
100	Salim	Mehmood	90
101	Abdul	Rasheed	90
102	Zia Ul	Hassan	90
103	S.M.	Ali	60
204	Hamad	Malik	70
205	Sohail	Aftab	110
206	Syed Wasim	Noor	110

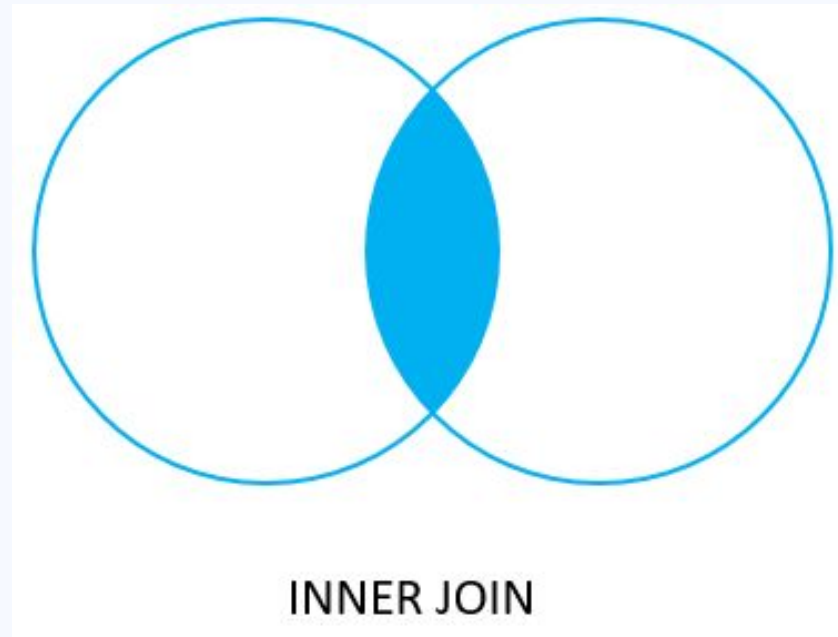
DEPARTMENT_ID	DEPARTMENT_NAME	LOCATION_ID
10	Administration	1700
20	Marketing	1800
30	Purchasing	1700
40	Human Resources	2400
50	Shipping	1500
60	IT	1400
70	Public Relations	2700
80	Sales	2500



EMPLOYEE_ID	DEPARTMENT_ID	DEPARTMENT_NAME
100	90	Executive
101	90	Executive
102	90	Executive
103	60	IT
204	70	Public Relations
205	110	Accounting
206	110	Accounting

Inner join

If there are matching values in the two tables, the records with the matching values are returned to form a set of rows in a temporary table by SQL inner join. The pictorial representation of SQL inner join is represented below:



SYNTAX

```
SELECT column_name(s)
FROM table1
INNER JOIN table2
ON table1.column_name = table2.column_name;
```

It is a good practice to include the table name when specifying columns in the SQL statement.

```
SELECT table1.column1,table1.column2,table2.column1,....
FROM table1
INNER JOIN table2
ON table1.matching_column = table2.matching_column;
```

Identify Mistake

Worksheet	Query Builder
	<pre>SELECT EMPLOYEES.EMPLOYEE_ID, DEPARTMENTS.DEPARTMENT_ID FROM employees INNER JOIN departments ON employees.employee_id = DEPARTMENTS.DEPARTMENT_ID;</pre>

Example 1

Orders Table

IDord	IDCust	Ordval
1	123	12
2	456	34
3	789	56
4	012	78

Customers Table

IDCust	Custname	Origin
456	ABC	INDIA
012	DEF	US
809	GHI	CHINA
134	JKL	GERMANY

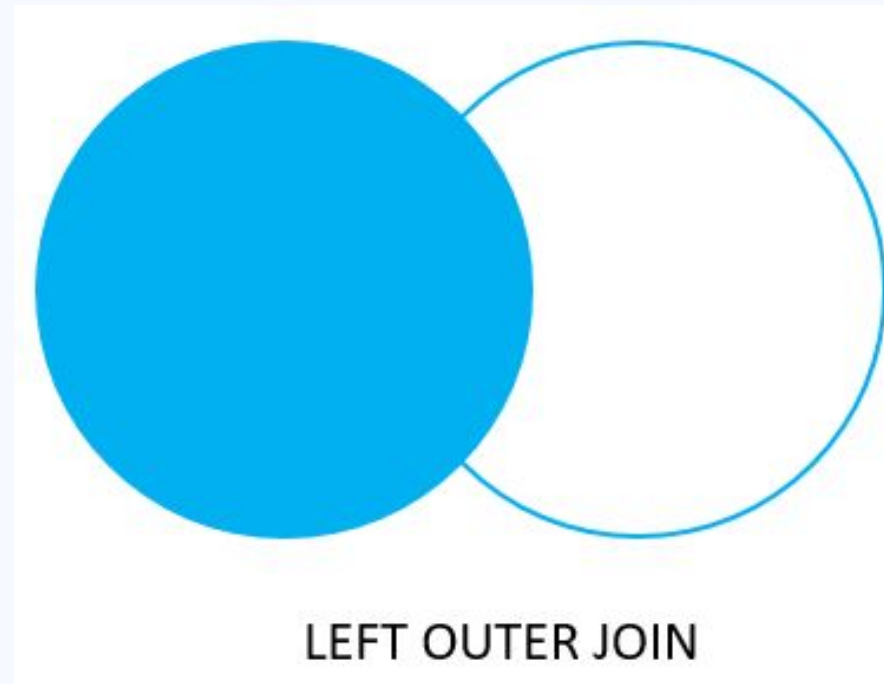
```
SELECT *  
FROM song  
JOIN artist  
    ON song.artist_id=artist.artist_id
```

song_id	song_title	artist_id	artist_id	artist_name
100	Dancing Queen	04	04	ABBA
101	Imagine	02	02	John Lennon
102	I Will Survive	01	01	Gloria Gaynor
104	Respect	03	03	Aretha Franklin
105	Mamma Mia	04	04	ABBA

Left join

A SQL LEFT should be used in cases when we want all the data from one table and only matching data from the other table.

The following Venn diagram explains how SQL left join works.



SYNTAX

```
SELECT table1.column1,table1.column2,table2.column1,....  
FROM table1  
LEFT JOIN table2  
ON table1.matching_column = table2.matching_column;
```

CONTD

Student

ROLL_NO	NAME	ADDRESS	PHONE	Age
1	HARSH	DELHI	XXXXXXXXXX	18
2	PRATIK	BIHAR	XXXXXXXXXX	19
3	RIYANKA	SILIGURI	XXXXXXXXXX	20
4	DEEP	RAMNAGAR	XXXXXXXXXX	18
5	SAPTARHI	KOLKATA	XXXXXXXXXX	19
6	DHANRAJ	BARABAJAR	XXXXXXXXXX	20
7	ROHIT	BALURGHAT	XXXXXXXXXX	18
8	NIRAJ	ALIPUR	XXXXXXXXXX	19

StudentCourse

COURSE_ID	ROLL_NO
1	1
2	2
2	3
3	4
1	5
4	9
5	10
4	11

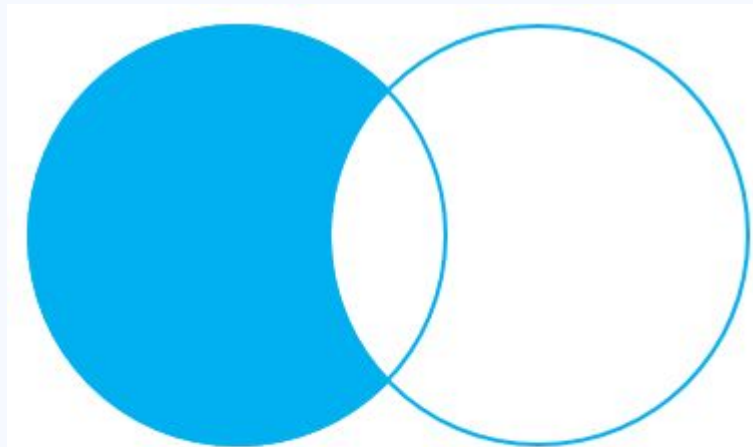
Example Queries(LEFT JOIN):

```
SELECT Student.NAME,StudentCourse.COURSE_ID  
FROM Student  
LEFT JOIN StudentCourse  
ON StudentCourse.ROLL_NO = Student.ROLL_NO;
```

Output:

NAME	COURSE_ID
HARSH	1
PRATIK	2
RIYANKA	2
DEEP	3
SAPTARHI	1
DHANRAJ	NULL
ROHIT	NULL
NIRAJ	NULL

Left outer join The following Venn diagram illustrates the left join with the exclusion of rows from the right table:



LEFT OUTER JOIN – only rows from the left table

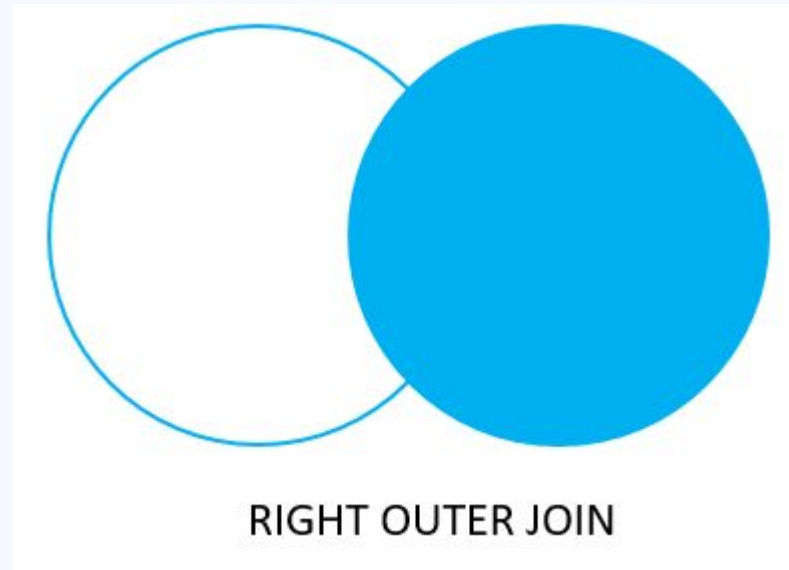
Right join

The right join or right outer join is a reversed version of the left join.

The right join makes a result set that contains all rows from the right table with the matching rows from the left table.

If there is no match, the left side will have nulls.

Venn diagram



SYNTAX

Syntax:

```
SELECT table1.column1,table1.column2,table2.column1,....  
FROM table1  
RIGHT JOIN table2  
ON table1.matching_column = table2.matching_column;
```

table1: First table.

table2: Second table

matching_column: Column common to both the tables.

Example Queries(RIGHT JOIN):

```
SELECT Student.NAME, StudentCourse.COURSE_ID  
FROM Student  
RIGHT JOIN StudentCourse  
ON StudentCourse.ROLL_NO = Student.ROLL_NO;
```

Output:

NAME	COURSE_ID
HARSH	1
PRATIK	2
RIYANKA	2
DEEP	3
SAPTARHI	1
NULL	4
NULL	5
NULL	4

